

OSSP – SKILL 2

Building a simple shell — the main loop: display a prompt, read a command, and exit on "exit".

Wrote the shell's main loop in `src/shell.c` using nano — it displays a `myshell>` prompt, reads a command with `scanf`, and stops when the user types `exit`.

```
GNU nano 8.7.1 src/shell.c
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];           // stores the command

    while (1) {                // main loop
        printf("myshell> ");    // show the prompt
        scanf("%s", input);     // read what the user types

        if (strcmp(input, "exit") == 0) { // strcmp is 0 when the words match
            printf("Exiting shell...\n"); // user typed exit -> stop the loop
            break;
        }

        printf("You typed: %s\n", input); // otherwise show the command
    }
    return 0;
}
```

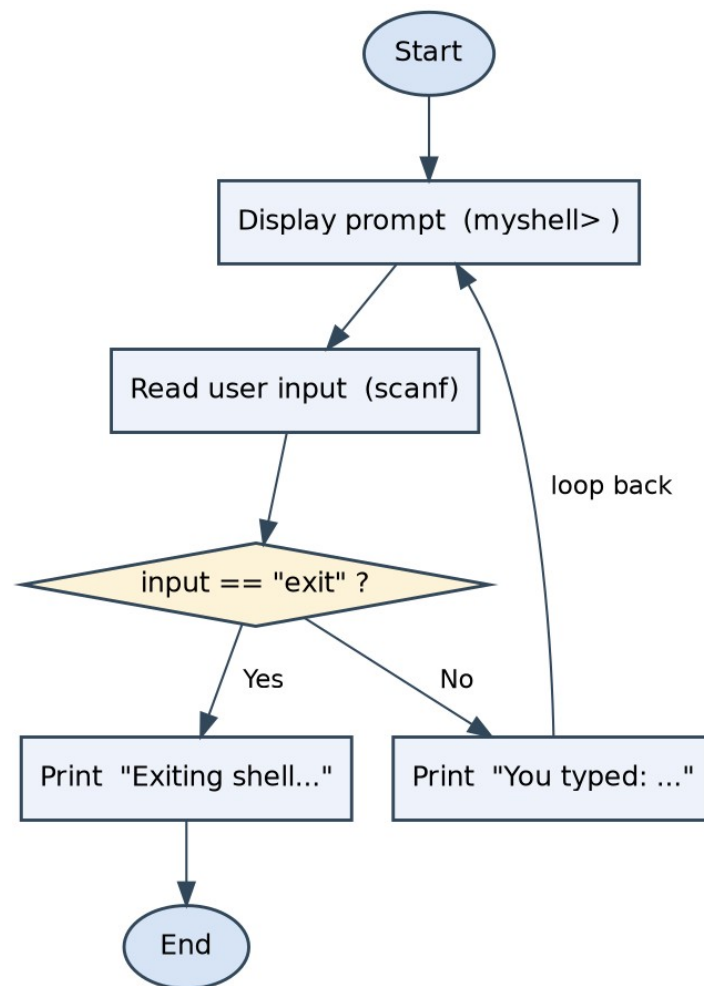
Compiled the program with `gcc src/shell.c` and started it with `./a.out` — the `myshell>` prompt appears, ready for input.

```
manu@Manasvi:~/os_lab/oss_p_shell$ nano src/shell.c
manu@Manasvi:~/os_lab/oss_p_shell$ nano src/shell.c
manu@Manasvi:~/os_lab/oss_p_shell$ gcc src/shell.c
manu@Manasvi:~/os_lab/oss_p_shell$ ./a.out
myshell> |
```

Tested the shell: typed `hello` and `date` (each is echoed back), then `exit`, which prints "Exiting shell..." and ends the loop.

```
manu@Manasvi:~/os_lab/oss_p_shell$ ./a.out
myshell> hello
You typed: hello
myshell> date
You typed: date
myshell> exit
Exiting shell...
manu@Manasvi:~/os_lab/oss_p_shell$ |
```

Control Flow Diagram



The shell repeats one loop: show the prompt, read the input, then either exit (if the command is exit) or echo the command and loop back.